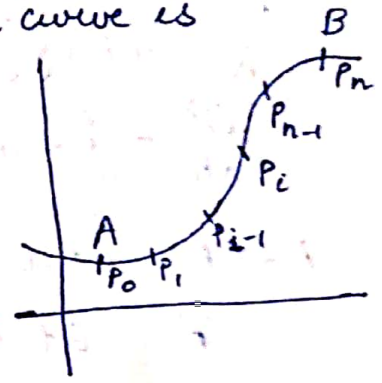


Line Integral

Any integral which is to be evaluated along a curve is called a line integral and is written as

$$\int_C \vec{F} \cdot d\vec{r} \quad \text{where } d\vec{r} = \hat{i}dx + \hat{j}dy + \hat{k}dz$$

$$\text{or } \int_C \vec{F} \cdot \frac{d\vec{r}}{dt} dt$$



If the path of integration is a closed curve we use the symbol $\oint_C \vec{F} \cdot d\vec{r}$

Let $\vec{F} = F_1\hat{i} + F_2\hat{j} + F_3\hat{k}$ and $d\vec{r} = \hat{i}dx + \hat{j}dy + \hat{k}dz$, then we have

$$\int_C \vec{F} \cdot d\vec{r} = \int_C (F_1 dx + F_2 dy + F_3 dz)$$

* If $\vec{F} = \vec{v}$, where v is the velocity of the fluid particle, then the line integral $\oint_C \vec{v} \cdot d\vec{r}$ is called the Circulation of the velocity around a closed curve C . Circulation is zero in a conservative field.

* If F is the gradient of some scalar point function ϕ , the line integral of $\vec{F}$ is independent of the path as follows:
Suppose $\vec{F} = \nabla\phi$ then $\int_A^B \vec{F} \cdot d\vec{r} = \int_A^B \nabla\phi \cdot d\vec{r} = [d\phi]_A^B = [\phi]_A^B = \phi(B) - \phi(A)$
In this case the line integral of $\vec{F}$ depends only on the end points A and B and not on the paths joining A & B .
 $\therefore$ work done $= \int_A^B \vec{F} \cdot d\vec{r} = \phi_B - \phi_A$.

Q. If $\vec{F} = 3xy\hat{i} - y^2\hat{j}$, evaluate $\int_C \vec{F} \cdot d\vec{r}$, where C is the arc of the parabola $y = 2x^2$ from $(0,0)$ to $(1,2)$

Solⁿ. Here $y = 2x^2$, let $x = t$ then $y = 2t^2$
At $(0,0)$, $x = 0$ and so $t = 0$
At $(1,2)$, $x = 1$ and so $t = 1$

If $\vec{r}$ is the position vector of any point (x, y) in C , then

$$\vec{r} = x\hat{i} + y\hat{j} + \cancel{z\hat{k}} = t\hat{i} + 2t^2\hat{j}$$

$$\frac{d\vec{r}}{dt} = \hat{i} + 4t\hat{j}$$

$$\text{Also } \vec{F} = (3 \cdot t \cdot 2t^2)\hat{i} - (2t^2)^2\hat{j}$$

$$\vec{F} = 6t^3\hat{i} - 4t^4\hat{j}$$

$$\begin{aligned} \therefore \int_C \vec{F} \cdot d\vec{r} &= \int_C \left(\vec{F} \cdot \frac{d\vec{r}}{dt} \right) dt = \int_0^1 (6t^3\hat{i} - 4t^4\hat{j}) \cdot (\hat{i} + 4t\hat{j}) dt \\ &= \int_0^1 (6t^3 - 16t^5) dt = \left[\frac{6t^4}{4} - \frac{16t^6}{6} \right]_0^1 = \frac{3}{2} - \frac{8}{3} = -\frac{7}{6} \end{aligned}$$

Q. Let the vector field be $\vec{F} = x^2\hat{i} - xy\hat{j}$. Find the line integral from the origin O to $P(1, 1)$.

(i) along a st. line OP

(ii) along the parabola $y^2 = x$

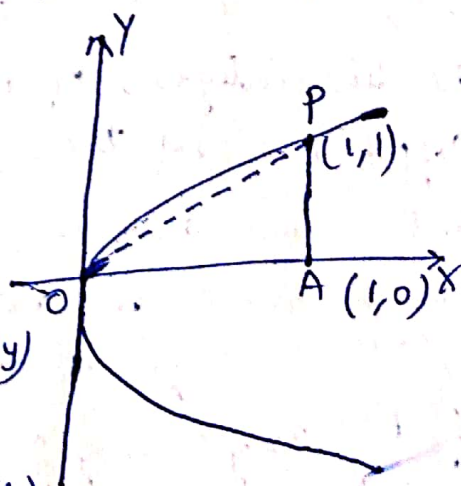
(iii) along the x -axis from O to $A(1, 0)$ and then parallel to y -axis from $A(1, 0)$ to $P(1, 1)$.

Solⁿ - Equation of the line OP is $y = x$
so that $dy = dx$

$$\therefore \int_{\text{line } OP} \vec{F} \cdot d\vec{r} = \int_{\text{line } OP} (x^2\hat{i} - xy\hat{j}) \cdot (\hat{i}dx + \hat{j}dy)$$

$$= \int_{\text{line } OP} (x^2 dx - xy dy) = \int_{\text{line } OP} (x^2 dx - x^2 dx)$$

$$= 0 \quad (\because y = x, xy = x^2)$$



(ii)

Equation of the parabola is $y^2 = x$, so that $2y dy = dx$
or $y dy = \frac{dx}{2}$

$$\begin{aligned} \therefore \int_{\text{Curve OP}} \vec{F} \cdot d\vec{r} &= \int_{\text{Curve OP}} (x^2 dx - xy dy) \\ &= \int_0^1 \left(x^2 - \frac{x}{2}\right) dx = \left[\frac{x^3}{3} - \frac{x^2}{4}\right]_0^1 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12} \end{aligned}$$

(iii) Here we integrate $\vec{F}$ from O to A along x -axis and from A to P parallel to y -axis.

$$\begin{aligned} \therefore \int_{\text{OAP}} \vec{F} \cdot d\vec{r} &= \int_{\text{OAP}} (x^2 dx - xy dy) = \int_{\text{OA}} x^2 dx - 0 + \int_{\text{AP}} (0 - 1y dy) \\ &= \left[\frac{x^3}{3}\right]_0^1 - \left[\frac{y^2}{2}\right]_0^1 = \frac{1}{3} - \frac{1}{2} = -\frac{1}{6} \end{aligned}$$

Q. Find the work done in moving a particle in a force field $\vec{F} = 3xy \hat{i} - 5z \hat{j} + 10x \hat{k}$ along the curve $x = t^2 + 1$, $y = 2t^2$, $z = t^3$ from $t = 0$ to $t = 2$

Solⁿ $\rightarrow$ Work done = $\int_C \vec{F} \cdot d\vec{r} = \int_C (3xy \hat{i} - 5z \hat{j} + 10x \hat{k}) \cdot (\hat{i} dx + \hat{j} dy + \hat{k} dz)$

$$= \int_C (3xy dx - 5z dy + 10x dz)$$

The curve C is $x = t^2 + 1$, $y = 2t^2$, $z = t^3$ from $t = 0$ to 2

$$\begin{aligned} \therefore \text{Work done} &= \int_0^2 3(t^2+1)(2t^2) dt - 5t^3(4t) dt + 10(t^2+1)3t^2 dt \\ &= \int_0^2 (12(t^5+t^3) - 20t^4 + 30(t^4+t^2)) dt \\ &= \int_0^2 (12t^5 + 10t^4 + 12t^3 + 30t^2) dt \\ &= \left[\frac{12t^6}{6} + \frac{10t^5}{5} + \frac{12t^4}{4} + \frac{30t^3}{3} \right]_0^2 = 16(8+4+3+5) = 320 \text{ unit} \end{aligned}$$

4.
Q. If $\vec{F} = xy\hat{i} - 3z\hat{j} + x^2\hat{k}$ and C is the curve $x=t^2, y=2t, z=t^3$ from $t=0$ to $t=1$, evaluate the line integral $\int_C \vec{F} \cdot d\vec{r}$

Solⁿ Since $x=t^2, y=2t$ and $z=t^3$, we have

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k} = t^2\hat{i} + 2t\hat{j} + t^3\hat{k}$$

$$\text{hence } d\vec{r} = (2t\hat{i} + 2\hat{j} + 3t^2\hat{k}) dt$$

$$\text{Also } \vec{F} = xy\hat{i} - 3z\hat{j} + x^2\hat{k} = 2t^3\hat{i} - t^3\hat{j} + t^4\hat{k}$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \int (2t^3\hat{i} - t^3\hat{j} + t^4\hat{k}) \cdot (2t\hat{i} + 2\hat{j} + 3t^2\hat{k}) dt$$

$$= \int [i(-3t^5 - 2t^4) + j(2t^5 - 6t^5) + k(4t^3 + 2t^4)] dt$$

$$= i\left(-\frac{3t^6}{6} - \frac{2t^5}{5}\right) + j\left(\frac{2t^6}{6} - t^6\right) + k\left(t^4 + \frac{2t^5}{5}\right)$$

$$\therefore \int_{t=0}^1 \vec{F} \cdot d\vec{r} = i\left(-\frac{3}{6} - \frac{2}{5}\right) + j\left(-\frac{2}{3}\right) + k\left(1 + \frac{2}{5}\right)$$

$$= -\frac{9\hat{i}}{10} - \frac{2\hat{j}}{3} + \frac{7\hat{k}}{5}$$

Q. A vector field is given by $\vec{F} = (\sin y)\hat{i} + x(1 + \cos y)\hat{j}$. Evaluate the line integral over the circular path given by $x^2 + y^2 = a^2, z=0$

Solⁿ The parametric equations of the circular path are

$$x = a \cos t, y = a \sin t, z = 0 \text{ where } t \text{ varies from } 0 \text{ to } 2\pi.$$

$$\text{or } dx = -a \sin t dt, dy = a \cos t dt$$

Since the particle moves in the xy plane ($z=0$),

we can take $\vec{r} = x\hat{i} + y\hat{j}$ so that $d\vec{r} = dx\hat{i} + dy\hat{j}$

$$\therefore \oint_C \vec{F} \cdot d\vec{r} = \oint_C [\sin y \hat{i} + x(1 + \cos y) \hat{j}] \cdot (dx\hat{i} + dy\hat{j})$$

$$= \oint_C [\sin y dx + x(1 + \cos y) dy] = \oint_C [\sin y dx + x \cos y dy + x dy]$$

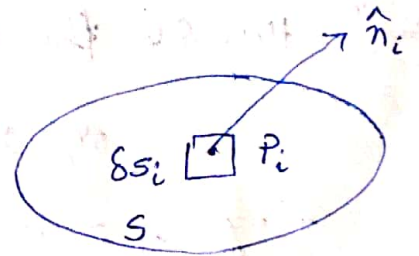
$$= \oint_C d(x \sin y) + \oint_C x dy = \int_0^{2\pi} d[a \cos t \sin(a \sin t)] + \int_0^{2\pi} a \cos t \cdot a \cos t dt$$

$$\begin{aligned}
 &= \left[a \cos t \sin(a \sin t) \right]_0^{2\pi} + a^2 \int_0^{2\pi} \cos^2 t \, dt \\
 &= \frac{a^2}{2} \int_0^{2\pi} (1 + \cos 2t) \, dt = \frac{a^2}{2} \left[t + \frac{\sin 2t}{2} \right]_0^{2\pi} = \frac{a^2}{2} (2\pi) = \pi a^2
 \end{aligned}$$

Surface Integral

Any integral which is to be evaluated over a surface is called a surface integral.

The normal surface integral of $\vec{F}(P)$ over S is denoted by $\iint_S \vec{F} \cdot \hat{n} \, ds = \iint_S \vec{F} \cdot d\vec{s}$



It is a measure of the flux of $\vec{F}$ over S .

Other types of surface integrals are $\iint_S \vec{F} \times d\vec{s}$, $\iint_S \phi \, d\vec{s}$

Let R be the orthogonal projection of S on the xy -plane.

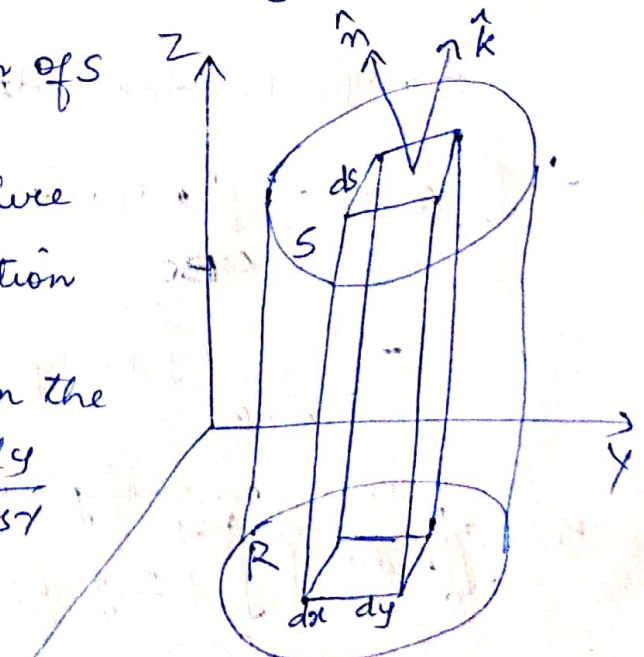
Let $\hat{n} = \cos \alpha \hat{i} + \cos \beta \hat{j} + \cos \gamma \hat{k}$, where $\cos \alpha$, $\cos \beta$ and $\cos \gamma$ are the direction cosines of $\hat{n}$.

Now, $dxdy = \text{projection of } ds \text{ on the } xy\text{-plane} = ds \cos \gamma \Rightarrow ds = \frac{dxdy}{\cos \gamma}$

Also $|\hat{k} \cdot \hat{n}| = \cos \gamma$

$$\therefore ds = \frac{dxdy}{|\hat{k} \cdot \hat{n}|}$$

$$\text{Hence } \iint_S \vec{F} \cdot \hat{n} \, ds = \iint_R \vec{F} \cdot \hat{n} \frac{dxdy}{|\hat{k} \cdot \hat{n}|}$$



$$\begin{aligned}
 &\text{or, } \iint_R \vec{F} \cdot \hat{n} \frac{dydz}{|\hat{i} \cdot \hat{n}|} \\
 &\text{or, } \iint_R \vec{F} \cdot \hat{n} \frac{dxdz}{|\hat{j} \cdot \hat{n}|}
 \end{aligned}$$

$$\text{Also } \iint_S \vec{F} \cdot \hat{n} \, ds = \iint_S \vec{F} \cdot d\vec{s}$$

where $d\vec{s}$ is the vector $(dydz\hat{i} + dzdx\hat{j} + dxdy\hat{k})$

If $\vec{F} = F_1\hat{i} + F_2\hat{j} + F_3\hat{k}$ then

$$\iint_S \vec{F} \cdot d\vec{s} = \iint_S F_1 dydz + F_2 dzdx + F_3 dxdy$$

Q. Evaluate $\iint_S \vec{A} \cdot d\vec{s}$ where $\vec{A} = x\hat{i} + (z^2 - zx)\hat{j} - xy\hat{k}$ and S is the surface of the triangle with vertices $(2,0,0)$, $(0,2,0)$, $(0,0,4)$

Solⁿ Here S is the plane surface $\frac{x}{2} + \frac{y}{2} + \frac{z}{2} = 1$ or,

or, $2x + 2y + z = 4$ bounded by the lines

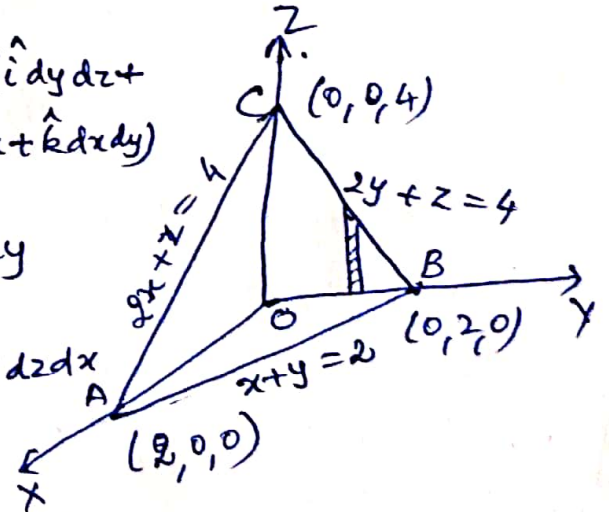
$x+y=2, z=0$; $2y+z=4, x=0$ and $2x+z=4, y=0$

Since $d\vec{s} = i dydz + j dzdx + k dxdy$, we have

$$\vec{A} \cdot d\vec{s} = [x\hat{i} + (z^2 - zx)\hat{j} - xy\hat{k}] \cdot (\hat{i} dydz + \hat{j} dzdx + \hat{k} dxdy)$$

$$= x dydz + (z^2 - zx) dzdx - xy dxdy$$

$$\therefore \iint_S \vec{A} \cdot d\vec{s} = \iint_{\triangle OBC} x dydz + \iint_{\triangle OAC} (z^2 - zx) dzdx - \iint_{\triangle OAB} xy dxdy$$



For points on the given surface, we have $x = 2 - y - \frac{1}{2}z$

Hence
$$\iint_S \vec{A} \cdot d\vec{s} = \int_0^2 \int_0^{4-2y} (2-y-\frac{1}{2}z) dz dy + \int_0^2 \int_0^{4-2x} (z^2 - zx) dz dx - \int_0^2 \int_0^{2-x} xy dy dx$$

$$= \int_0^2 \left[2z - yz - \frac{z^2}{4} \right]_0^{4-2y} dy + \int_0^2 \left[\frac{z^3}{3} - \frac{xz^2}{2} \right]_0^{4-2x} dx - \int_0^2 \left[\frac{xy^2}{2} \right]_0^{2-x} dx$$

$$= \int_0^2 (2-y)^2 dy + \frac{8}{3} \int_0^2 (2-x)^3 dx - \frac{5}{2} \int_0^2 x(2-x)^2 dx$$

$$= \frac{8}{3} + \frac{32}{3} - \frac{10}{3} = 10$$

Q. Evaluate $\iint_S \vec{A} \cdot \hat{n} \, ds$ where $\vec{A} = (x+y^2)\hat{i} - 2xz\hat{j} + 2yz\hat{k}$ and S is the surface of the plane $2x+y+2z=6$ in the first octant.

Solⁿ: A vector normal to the surface S is given by

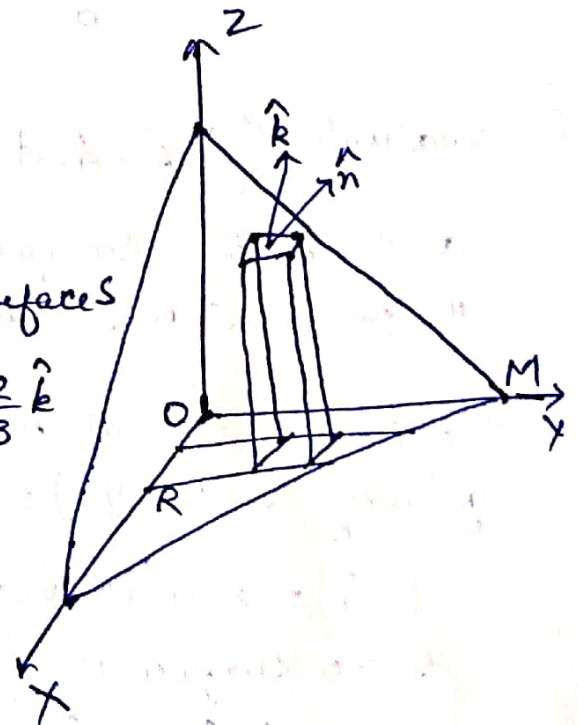
$$\nabla (2x+y+2z) = 2\hat{i} + \hat{j} + 2\hat{k}$$

$\therefore \hat{n}$ = a unit vector normal to surface S

$$= \frac{2\hat{i} + \hat{j} + 2\hat{k}}{\sqrt{2^2 + 1^2 + 2^2}} = \frac{2}{3}\hat{i} + \frac{1}{3}\hat{j} + \frac{2}{3}\hat{k}$$

$$\hat{k} \cdot \hat{n} = \hat{k} \cdot \left(\frac{2}{3}\hat{i} + \frac{1}{3}\hat{j} + \frac{2}{3}\hat{k} \right) = \frac{2}{3}$$

$$\therefore \iint_S \vec{A} \cdot \hat{n} \, ds = \iint_R \vec{A} \cdot \hat{n} \frac{dxdy}{|\hat{k} \cdot \hat{n}|}$$



where R is the projection of S , i.e. triangle LMN on the xy plane. The region R , i.e. triangle OLM is bounded by x -axis, y -axis and the line $2x+y=6$, $z=0$

$$\text{Now, } \vec{A} \cdot \hat{n} = [(x+y^2)\hat{i} - 2xz\hat{j} + 2yz\hat{k}] \cdot \left(\frac{2}{3}\hat{i} + \frac{1}{3}\hat{j} + \frac{2}{3}\hat{k} \right)$$

$$= \frac{2}{3}(x+y^2) - \frac{2}{3}xz + \frac{4}{3}yz = \frac{2}{3}y^2 + \frac{4}{3}yz$$

$$= \frac{2}{3}y^2 + \frac{4}{3}y \left(\frac{6-2x-y}{2} \right) \quad (\because \text{on the plane } 2x+y+2z=6, z=\frac{6-2x-y}{2})$$

$$= \frac{2}{3}y(y+6-2x-y) = \frac{4}{3}y(3-x)$$

Hence $\iint_S \vec{A} \cdot \hat{n} ds = \iint_R \vec{A} \cdot \hat{n} \frac{dx dy}{|\hat{k} \cdot \hat{n}|}$

$$= \iint_R \frac{4}{3} y(3-x) \cdot \frac{3}{2} dx dy = \int_0^3 \int_0^{6-2x} 2y(3-x) dy dx$$

$$= \int_0^3 2(3-x) \cdot \left[\frac{y^2}{2} \right]_0^{6-2x} dx = \int_0^3 (3-x)(6-2x)^2 dx$$

$$= 4 \int_0^3 (3-x)^3 dx = 4 \cdot \left[\frac{(3-x)^4}{4(-1)} \right]_0^3 = -(0-81) = \underline{\underline{81}}$$

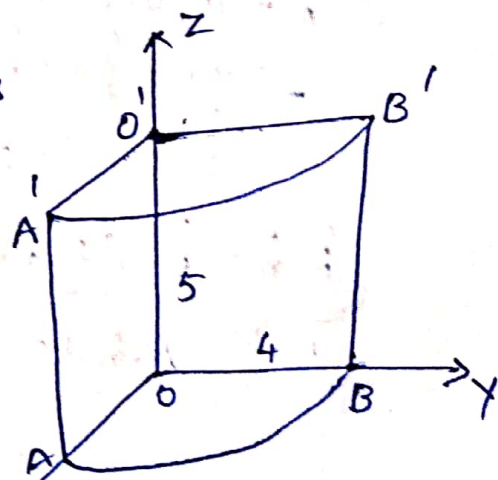
Q. evaluate $\iint_S \vec{A} \cdot \hat{n} ds$; where $\vec{A} = z\hat{i} + x\hat{j} - 3y^2z\hat{k}$ and S is the surface of the cylinder $x^2 + y^2 = 16$ included in the first octant between $z=0$ and $z=5$.

Solⁿ, A vector normal to the surface S is given by $\nabla(x^2 + y^2) = 2x\hat{i} + 2y\hat{j}$

$\therefore \hat{n}$ = a unit vector normal to the surface S

$$= \frac{2x\hat{i} + 2y\hat{j}}{\sqrt{(2x)^2 + (2y)^2}} = \frac{x\hat{i} + y\hat{j}}{\sqrt{x^2 + y^2}} = \frac{x\hat{i} + y\hat{j}}{4}$$

($\because$ on the surface of cylinder $x^2 + y^2 = 16$)



Let R be the projection of S on yz -plane, then

$$\iint_S \vec{A} \cdot \hat{n} ds = \iint_R \vec{A} \cdot \hat{n} \frac{dy dz}{|\hat{i} \cdot \hat{n}|}$$

The region R is $OBB'O'$ enclosed by $y=0$ to $y=4$ and $z=0$ to $z=5$

$$\text{Now, } \hat{i} \cdot \hat{n} = \hat{i} \cdot \left(\frac{1}{4}x\hat{i} + \frac{1}{4}y\hat{j} \right) = \frac{1}{4}x$$

$$\begin{aligned}\vec{A} \cdot \hat{n} &= (z\hat{i} + x\hat{j} - 3y^2z\hat{k}) \cdot \left(\frac{1}{4}x\hat{i} + \frac{1}{4}y\hat{j}\right) \\ &= \frac{1}{4}zx + \frac{1}{4}xy = \frac{1}{4}x(y+z)\end{aligned}$$

$$\begin{aligned}\text{Hence } \iint_S \vec{A} \cdot \hat{n} \, ds &= \iint_R \vec{A} \cdot \hat{n} \frac{dy \, dz}{|\hat{i} \cdot \hat{n}|} = \iint_R \frac{1}{4}x(y+z) \frac{dy \, dz}{\frac{1}{4}x} \\ &= \int_0^5 \int_0^4 (y+z) \, dy \, dz \\ &= \int_0^5 \left[\frac{y^2}{2} + zy \right]_0^4 \, dz = \int_0^5 (8 + 4z) \, dz \\ &= \left[8z + 2z^2 \right]_0^5 = 40 + 50 = 90\end{aligned}$$

Volume Integral

An integral which is evaluated over a volume is called volume integral.

If V is a volume bounded by a surface S , then the triple integrals: $\iiint_V \phi \, dV$ and $\iiint_V \vec{F} \, dV$ are called volume integrals.

The first of these is a scalar and the second is a vector.

If we subdivide the volume V into small cuboids by drawing planes parallel to the coordinate planes, then $dV = dx \, dy \, dz$

$$\therefore \iiint_V \phi \, dV = \iiint_V \phi(x, y, z) \, dx \, dy \, dz$$

If $\vec{F} = F_1\hat{i} + F_2\hat{j} + F_3\hat{k}$, then

$$\begin{aligned}\iiint_V \vec{F} \, dV &= \hat{i} \iiint_V F_1(x, y, z) \, dx \, dy \, dz + \hat{j} \iiint_V F_2(x, y, z) \, dx \, dy \, dz \\ &\quad + \hat{k} \iiint_V F_3(x, y, z) \, dx \, dy \, dz.\end{aligned}$$

Q. If $\vec{F} = (2x^2 - 3z)\hat{i} - 2xy\hat{j} - 4x\hat{k}$, then evaluate $\iiint_V \nabla \cdot \vec{F} dv$, where V is bounded by the plane $x=0, y=0, z=0, 2x+2y+z=4$

Solⁿ $\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(2x^2 - 3z) + \frac{\partial}{\partial y}(-2xy) + \frac{\partial}{\partial z}(-4x)$
 $= 2x$

$$\therefore \iiint_V \nabla \cdot \vec{F} dv = \iiint_V 2x dx dy dz$$

$\begin{cases} 2x+2y+z=4 \\ z=4-2x-2y \\ \text{when } z=0 \\ y=2-x \end{cases}$

$$= \int_0^2 \int_0^{2-x} \int_0^{4-2x-2y} 2x dz dy dx$$

$$= \int_0^2 \int_0^{2-x} 2x [z]_0^{4-2x-2y} dy dx$$

$$= \int_0^2 \int_0^{2-x} 2x(4-2x-2y) dy dx$$

$$= \int_0^2 \left\{ \int_0^{2-x} (8x - 4x^2 - 4xy) dy \right\} dx$$

$$= \int_0^2 \left[8xy - 4x^2y - 2xy^2 \right]_0^{2-x} dx$$

$$= \int_0^2 [4x(2-x)^2 - 2x(2-x)^2] dx$$

$$= \int_0^2 2x(2-x)^2 dx$$

$$= 2 \int_0^2 (4x - 4x^2 + x^3) dx$$

$$= 2 \left[2x^2 - \frac{4x^3}{3} + \frac{x^4}{4} \right]_0^2 = 2 \left(8 - \frac{32}{3} + 4 \right) = \frac{8}{3}$$

Q. If $A = 2xz\hat{i} - x\hat{j} + y^2\hat{k}$, evaluate $\iiint_V \vec{A} \, dv$, where V is the region bounded by the surface $x=0, y=0, x=2, y=6, z=x^2, z=4$

Solⁿ $\rightarrow \iiint_V \vec{A} \, dv = \iiint_V (2xz\hat{i} - x\hat{j} + y^2\hat{k}) \, dx \, dy \, dz$

$$= \int_0^2 \int_0^6 \int_{x^2}^4 (2xz\hat{i} - x\hat{j} + y^2\hat{k}) \, dz \, dy \, dx$$

$$= \hat{i} \int_0^2 \int_0^6 \int_{x^2}^4 2xz \, dz \, dy \, dx - \hat{j} \int_0^2 \int_0^6 \int_{x^2}^4 x \, dz \, dy \, dx +$$

$$+ \hat{k} \int_0^2 \int_0^6 \int_{x^2}^4 y^2 \, dz \, dy \, dx$$

$$= \hat{i} \int_0^2 \int_0^6 \left[xz^2 \right]_{x^2}^4 \, dy \, dx - \hat{j} \int_0^2 \int_0^6 \left[xz \right]_{x^2}^4 \, dy \, dx + \hat{k} \int_0^2 \int_0^6 \left[y^2 z \right]_{x^2}^4 \, dy \, dx$$

$$= \hat{i} \int_0^2 \int_0^6 (16x - x^5) \, dy \, dx - \hat{j} \int_0^2 \int_0^6 (4x - x^3) \, dy \, dx + \hat{k} \int_0^2 \int_0^6 y^2 (4 - x^2) \, dy \, dx$$

$$= \hat{i} \int_0^2 (16x - x^5) \left[y \right]_0^6 \, dx - \hat{j} \int_0^2 (4x - x^3) \left[y \right]_0^6 \, dx + \hat{k} \int_0^2 (4 - x^2) \left[\frac{y^3}{3} \right]_0^6 \, dx$$

$$= 6\hat{i} \int_0^2 (16x - x^5) \, dx - 6\hat{j} \int_0^2 (4x - x^3) \, dx + 72\hat{k} \int_0^2 (4 - x^2) \, dx$$

$$= 6\hat{i} \int_0^2 (16x - x^5) \, dx$$

$$= 6\hat{i} \left[8x^2 - \frac{x^6}{6} \right]_0^2 - 6\hat{j} \left[2x^2 - \frac{x^4}{4} \right]_0^2 + 72\hat{k} \left[4x - \frac{x^3}{3} \right]_0^2$$

$$= 128\hat{i} - 24\hat{j} + 384\hat{k}$$